# gGA Dimer Scans: $U/t$ Dependence at $\Delta V = 0$ and $\Delta V = 2$

(Ghost Multiplicities $N_g = 0$ and $N_g = 2$, Direct-Minimization Solver)

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## 1 The $N_g = 0$ case (ordinary Gutzwiller)

At $\Delta V = 0$ the Gutzwiller equations of the dimer have a closed solution (`gGut_notes.pdf`, Sec. III D). With $V = -tR$ and $\lambda = \lambda^c = 0$ the embedding ground state gives, for $U \le U_c = 8t$ (Brinkman–Rice),

$$Z = R^2 = 1 - \Big(\frac{U}{8t}\Big)^2, \qquad E = -2t - \frac{U}{2} - \frac{U^2}{32t}, \qquad \langle n_\uparrow n_\downarrow \rangle = \frac{1}{4} - \frac{U}{32t}, \qquad P_{\rm ion} = \frac{1}{2} - \frac{U}{16t}, \quad (1)$$

and for $U \ge 8t$ only $R = 0$ exists: $Z = 0$, $E = -U$, $P_{\rm ion} = 0$. The quasiparticle Hamiltonian is $-tZ\,\sigma_x$, with levels $\pm tZ$ and orbitals $(1, \pm 1)/\sqrt{2}$ on the two states; each level carries the weight $2Z$, so the spectral function consists of two peaks, at $\pm tZ$ with weight $2Z$, that move to $\omega = 0$ and vanish at $U_c$ (the gap $2tZ$ closes continuously). Here $U_c$ is the value where the spectral function disappears. The closed forms agree with the numerical $N_g = 0$ solution of the code to $1.4 \times 10^{-7}$ in $Z$ and $10^{-8}$ in $E$ and $\langle n_\uparrow n_\downarrow \rangle$ (script `ng0_analysis.py`).

At $\Delta V \neq 0$ particle–hole symmetry is broken, so $\lambda \neq 0$ and the embedding problem is $3\times3$ in the states $S = (a^\dagger_\uparrow d^\dagger_\downarrow - a^\dagger_\downarrow d^\dagger_\uparrow)/\sqrt{2}$, $D = a^\dagger_\uparrow a^\dagger_\downarrow$, $E = d^\dagger_\uparrow d^\dagger_\downarrow$ of the physical ($a$) and bath ($d$) orbitals. The site exchange combined with the particle–hole transformation relates the two fragments, $\phi_1 = (s, \epsilon, \delta)$ for $\phi_0 = (s, \delta, \epsilon)$ in the basis $(S, D, E)$, so $Z_0 = Z_1$ and the problem has two variables, $c = \delta^2 + \epsilon^2$ and $m = \delta^2 - \epsilon^2$. With $r = \sqrt{c^2 - m^2}$:

$$E = -U(1-c) - 2\Delta V\, m - 4t\,\frac{(1-c)(c+r)}{\sqrt{1-m^2}}, \qquad Z = \frac{2(1-c)(c+r)}{1-m^2}, \qquad P_{\rm ion} = c, \qquad n = 1+m, \tag{2}$$

where $n$ is the number of electrons on the favored site; the quasiparticle levels are $\pm tZ/\sqrt{1-m^2}$ with weight $2Z$ each ($\lambda = tZm/\sqrt{1-m^2}$), and at $\Delta V = 0$ ($m = 0$) all of this reduces to the formulas above. The stationarity conditions, $U\sqrt{1-m^2}\,r = 4t(c+r)(1-c-r)$ and $\Delta V(1-m^2)^{3/2} r = 2t(1-c)\,m\,(1-c^2-rc)$, are solved for $(c, m)$; they reproduce the numerical $N_g = 0$ solution of the code (`scan_ng0/`) to $8\times10^{-8}$ in $E$, $4\times10^{-7}$ in $Z$ and $5\times10^{-8}$ in $P_{\rm ion}$ (script `ng0_dV_analytic.py`). $Z$ vanishes for $c, m \to 0$ with $q = r/c$ fixed, where the two conditions give $U_c = 4t(1+q)/q$ and $\Delta V = 2t\sqrt{1-q^2}/q$, i.e.

$$U_c = 4t + 2\sqrt{4t^2 + \Delta V^2}, \tag{3}$$

which is $8t$ at $\Delta V = 0$, $9.66\,t$ at $\Delta V = 2$ and grows as $2\Delta V$ for large $\Delta V$.

**Quasiparticle orbitals.** In the basis of the two states $H_{qp} = -\left(\begin{smallmatrix} \lambda & tZ \\ tZ & -\lambda \end{smallmatrix}\right)$ with $\lambda = tZ\,m/\sqrt{1-m^2}$, so the levels are $\mp E_q$, $E_q = tZ/\sqrt{1-m^2}$, and the orbitals (up to a global sign, chosen as that of the $N_g = 2$ HOMO and LUMO on the two states) are

$$|\mathrm{occ}\rangle = \left(\sqrt{\tfrac{1+m}{2}},\ \sqrt{\tfrac{1-m}{2}}\right), \qquad |\mathrm{virt}\rangle = \left(\sqrt{\tfrac{1-m}{2}},\ -\sqrt{\tfrac{1+m}{2}}\right), \tag{4}$$

i.e. the occupied orbital has the density $(1+m)/2$ on the favored state (0.945, 0.759 and 0.549 at $U/t = 0.1$, 4 and 8 for $\Delta V = 2$). At $\Delta V = 0$, $m = 0$ and the orbitals are $(1, \pm1)/\sqrt{2}$. Two limits are explicit: for $U \to 0$, $m \to \Delta V/\sqrt{t^2 + \Delta V^2}$, the non-interacting value (0.894 for $\Delta V = 2$; the solution gives 0.894 at $U/t = 0.01$); at $U_c$, $m \to 0$ and $Z \to 0$, so the orbitals become $(1, \pm1)/\sqrt{2}$ with levels at zero. For general $(U, \Delta V)$, $m$ is not an explicit function of $(U, \Delta V)$; instead $U$ and $\Delta V$ are explicit functions of $(c, m)$,

$$U = \frac{4t\,(c+r)(1-c-r)}{\sqrt{1-m^2}\,r}, \qquad \Delta V = \frac{2t\,(1-c)\,m\,(1-c^2-rc)}{(1-m^2)^{3/2}\,r}, \qquad r = \sqrt{c^2-m^2}, \tag{5}$$

which is the stationarity condition solved for $(U, \Delta V)$; for given $(U, \Delta V)$ the pair $(c, m)$ is its root (script `ng0_analytic.py`; the round trip $(U, \Delta V) \to (c, m) \to (U, \Delta V)$ closes to $10^{-6}$).
In summary, $m = \delta^2 - \epsilon^2$ (half the difference of the electron numbers of the two states) and the amplitudes $\delta^2 = (c+m)/2$, $\epsilon^2 = (c-m)/2$ are known in closed form only at $\Delta V = 0$ ($m = 0$, $\delta = \epsilon = \sqrt{1/4 - U/32t}$ for $U \leq 8t$), in the limit $U \to 0$ (above), and as $U \to U_c$ where $\delta, \epsilon \to 0$ with $m/c \to \Delta V/\sqrt{4t^2 + \Delta V^2}$; for all other $(U, \Delta V)$ they are given parametrically by the two equations above and obtained numerically from them. The critical $U_c/t$ is in closed form for all $\Delta V$.

Table 1 gives the main quantities.

| | $\Delta V = 0$ (closed form) | | | $\Delta V = 2$ (two-variable reduction) | | |
|---|---|---|---|---|---|---|
| $U/t$ | $Z$ | $E_{\mathrm{var}}/t$ | $P_{\mathrm{ion}}$ | $Z$ | $E_{\mathrm{var}}/t$ | $P_{\mathrm{ion}}$ |
| 2 | 0.938 | −3.125 | 0.375 | 0.993 | −4.774 | 0.781 |
| 4 | 0.750 | −4.500 | 0.250 | 0.926 | −5.422 | 0.558 |
| 6 | 0.438 | −6.125 | 0.125 | 0.724 | −6.549 | 0.321 |
| 7 | 0.234 | −7.031 | 0.062 | 0.567 | −7.280 | 0.220 |
| 8 | 0 | −8.000 | 0 | 0.378 | −8.105 | 0.130 |
| 10 | 0 | −10.000 | 0 | 0 | −10.000 | 0 |

Table 1: $N_g = 0$: renormalization factor, energy and ionic weight.

## 2 Total energy and error vs. exact

Figures 1 and 2 show the variational energy $E_{\mathrm{var}}/t$ of the $N_g = 0$ and $N_g = 2$ solutions together with the exact (ED) energy, and the error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ on a logarithmic scale, for $\Delta V = 0$ and $\Delta V = 2$ respectively. For $\Delta V = 2$ the plotted $N_g = 2$ energies are the cross-checked ("confirmed") solutions; the $N_g = 0$ energies are the closed form of Sec. 1 at $\Delta V = 0$ and its two-variable reduction at $\Delta V = 2$. The two curves agree to $10^{-3}$ up to $U/t = 2.3$ ($\Delta V = 0$) and 4.3 ($\Delta V = 2$); $N_g = 0$ is higher by at most 0.13 (at $U/t = 7.7$ and 9.4).

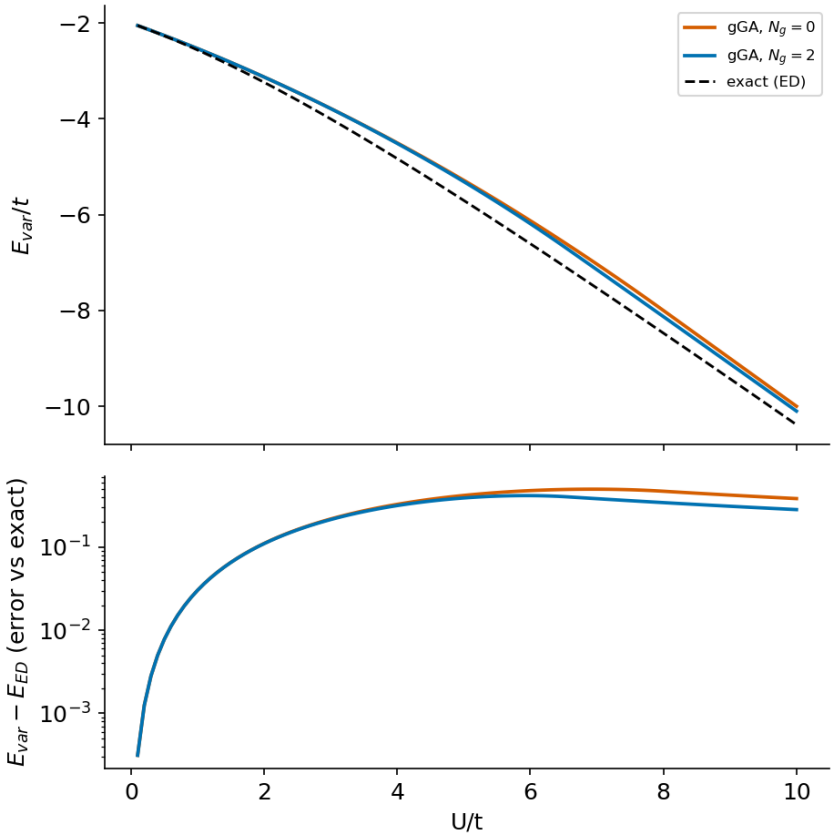



Figure 1: $\Delta V = 0$: $E_{\mathrm{var}}/t$ (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. $U/t$ for $N_g = 0$ and $N_g = 2$.

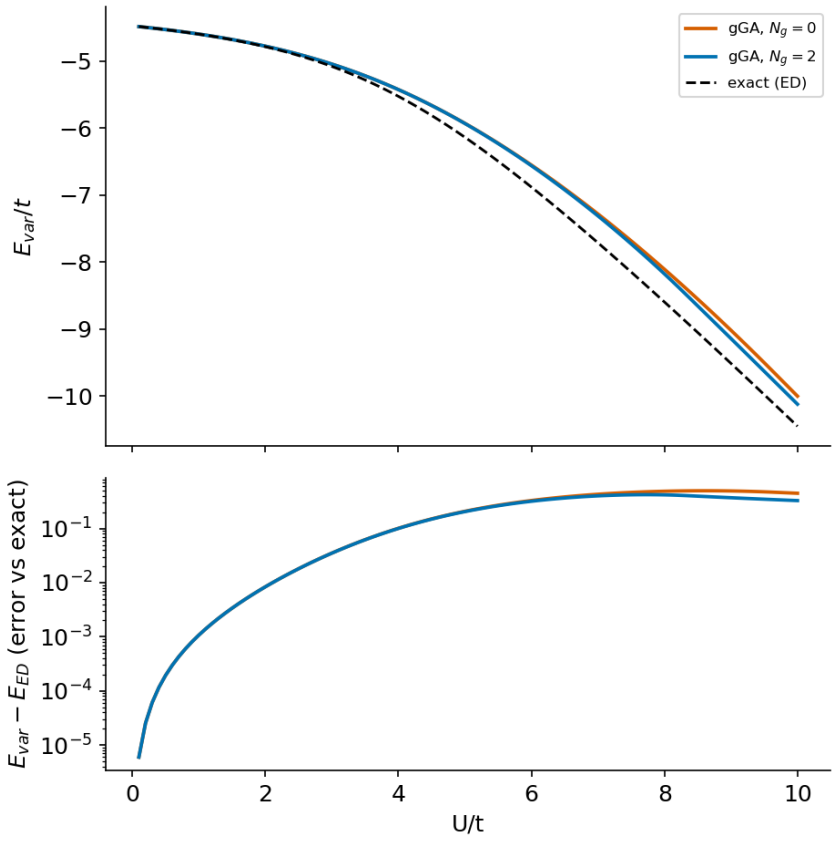
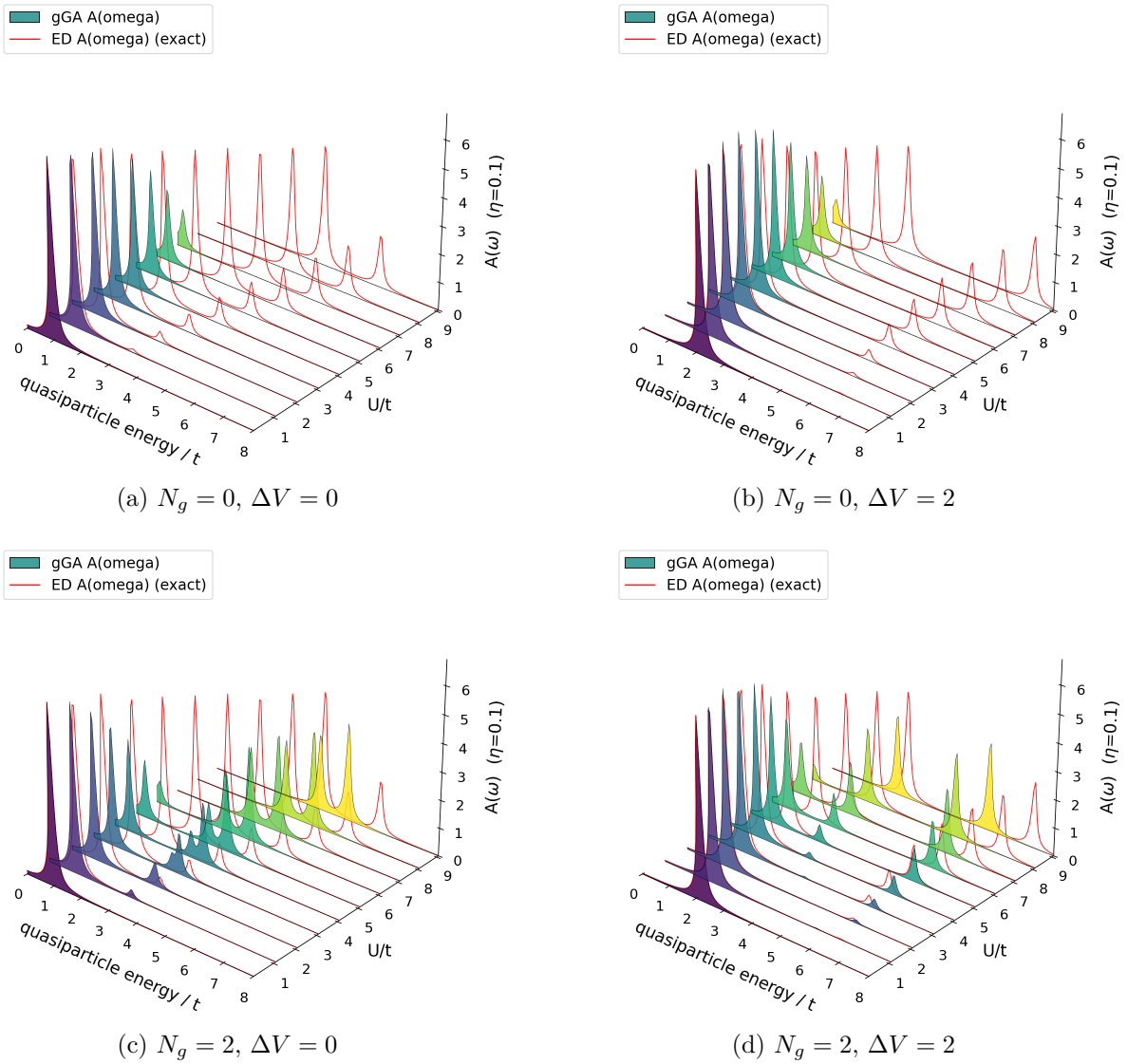



Figure 2: $\Delta V = 2$: $E_{\mathrm{var}}/t$ (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. $U/t$ for $N_g = 0$ and $N_g = 2$.

# 3 Spectra and spectral-function movies

## 3.1 3D view

Figure 3 shows, for each $(N_g, \Delta V)$ combination, the gGA spectral function $A(\omega)$ (filled ridgelines, built from the quasiparticle poles and weights of the converged solution, broadening $\eta = 0.1$) as a function of $U/t$, together with the exact (ED) spectral function (red lines). Rows are $N_g = 0$ and $N_g = 2$, columns are $\Delta V = 0$ and $\Delta V = 2$. The $N_g = 2$, $\Delta V = 2$ panel uses the cross-checked solutions.




(a) $N_g = 0$, $\Delta V = 0$

(b) $N_g = 0$, $\Delta V = 2$

(c) $N_g = 2$, $\Delta V = 0$

(d) $N_g = 2$, $\Delta V = 2$

Figure 3: gGA spectral function $A(\omega)$ (filled) vs. ED (red) as a function of $U/t$ for $N_g = 0$ (top) and $N_g = 2$ (bottom), at $\Delta V = 0$ (left) and $\Delta V = 2$ (right). Only $\omega \geq 0$ is shown: $A(\omega) = A(-\omega)$ in both gGA and ED; the $N_g = 0$ panels are the analytic solution. Scripts: `plot_eqp_3d_ng0.py`, `plot_eqp_3d_pos.py`.

**Analysis.** The number of quasiparticle branches is $2(1 + N_g)$: two for $N_g = 0$ and six for $N_g = 2$. In all four panels the gGA peaks coincide with the ED ones at small $U/t$ and move away from them as $U/t$ grows, consistent with the error in the total energy growing with $U/t$ (Sec. 2). For $N_g = 0$ there is a single positive peak, at $tZ$ with weight $2Z$ at $\Delta V = 0$, which goes to $\omega = 0$ and vanishes at $U_c/t = 8.00$ (9.66 at $\Delta V = 2$): the spectral function disappears. For $N_g = 2$ the extra branches give the outer peaks, and the inner peak vanishes at the lower $U_c/t = 6.56$ (8.33 at $\Delta V = 2$). The effect of $\Delta V$ is visible for both $N_g$: at $\Delta V = 2$ the branches are well separated at small $U/t$ and bunch together as $U/t$ increases, with a rearrangement for $N_g = 2$ between $U/t = 0.6$ and 0.7 (Sec. 5).

### 3.2 Spectral function movies

- $N_g = 0$, $\Delta V = 0$: spectral-function movie
- $N_g = 0$, $\Delta V = 2$: spectral-function movie
- $N_g = 2$, $\Delta V = 0$: spectral-function movie
- $N_g = 2$, $\Delta V = 2$: spectral-function movie

The $N_g = 0$ movies are drawn from the analytic solution of Sec. 1. (Links open the movie with the PDF viewer's associated application, if the viewer supports `run:` links.)

**Quasiparticle-weight movies.** The same scans are shown level by level: for each $U/t$ the eigenvalues $\epsilon_n^{qp}$ of $H_{qp}$ with the weight carried by each level (summed over both sites and spins, so the total weight is that of $A(\omega)$), and the ED poles as red open circles (linear and logarithmic scales; script `make_qp_weights_video.py`). A level with weight $\geq 10^{-3}$ is a peak (levels without weight are light gray). A peak keeps its color along $U/t$ (consecutive $U/t$ points are matched by closeness in energy and weight), and the colors **blue**, **orange**, **green** follow the order of increasing energy at the $U/t$ where all the peaks coexist.

- $N_g = 0$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 0$, $\Delta V = 2$: quasiparticle-weight movie
- $N_g = 2$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 2$, $\Delta V = 2$: quasiparticle-weight movie

**Weight transfer.** Positive-energy side. Each peak is written as (weight, energy) in the color of its peak in the movies (weights summed over sites and spins, total 2.0; ED poles in red). At $\Delta V = 0$ the peaks are the blue one, which carries almost all the weight at small $U/t$, and the orange and green ones, which are born at $U/t \simeq 0.5$ near $\epsilon = 3$. As $U/t$ grows the blue peak loses weight to the orange and green ones while its energy goes to zero, and both vanish at a critical $U_c/t$ (ED has no such event; the values come from a linear extrapolation of the last grid points); the orange and green peaks continue, with the same colors, above $U_c$.

- $\Delta V = 0$, $U/t \lesssim 1$ ($U/t = 1$): (1.92, 0.97) carries $\geq 96\%$; (0.02, 2.99) and (0.06, 3.02) hold the rest. ED: (1.97, 1.06), (0.03, 3.06).
- $\Delta V = 0$, $1 \lesssim U/t < U_c/t \simeq 6.56$: the blue peak loses its weight and the other two gain it. $U/t = 4$: (1.08, 0.57), (0.27, 2.83), (0.66, 3.26); $U/t = 6$: (0.27, 0.14), (0.66, 2.56), (1.05, 3.41); $U/t = 6.5$: (0.03, 0.02), (0.80, 2.46), (1.13, 3.43). ED at $U/t = 4$: (1.71, 1.83), (0.29, 3.83).
- $\Delta V = 0$, $U/t > U_c/t$: only the orange and green peaks. $U/t = 7$: (0.83, 2.69), (1.13, 3.67); $U/t = 10$: (0.89, 4.29), (1.10, 5.29). ED at $U/t = 10$: (1.37, 4.39), (0.63, 6.39).

At $\Delta V = 2$ the blue peak is again the main one; the orange peak is the middle one (born at $U/t \simeq 1.5$–$3$) and the green one the upper one (born at $U/t \simeq 1$).

- $\Delta V = 2$, $U/t \lesssim 3$ ($U/t = 3$): (1.89, 1.25), (0.015, 3.59), (0.10, 5.99); ED: (1.89, 1.34), (0.11, 5.81).
- $\Delta V = 2$, $3 \lesssim U/t < U_c/t \simeq 8.3$: the blue peak loses weight and the other two gain it. $U/t = 5$: (1.46, 0.81), (0.18, 2.57), (0.36, 5.90); $U/t = 6$: (1.10, 0.59), (0.37, 2.33), (0.53, 5.83); $U/t = 8$: (0.18, 0.09), (0.84, 2.08), (0.95, 5.28). The blue peak has weight 0.02 at $U/t = 8.3$ and $10^{-3}$ at 8.5 (it stays at $\epsilon \lesssim 10^{-3}$).
- $\Delta V = 2$, $U/t > U_c/t$: only the orange and green peaks. $U/t = 9$: (0.94, 2.38), (1.02, 5.66); $U/t = 10$: (0.96, 2.88), (1.02, 6.36). ED at $U/t = 10$: (1.22, 3.21), (0.78, 7.68).

Above $U_c$ the two weights approach each other but are not equal up to $U/t = 10$; gGA is closer to equal sharing than ED, i.e. it puts too much weight on the upper level.

### 3.3 Quasiparticle orbitals and density: the gap closing at $U_c$

Two more movies per case show the eigenorbitals of $H_{qp}$ and the quasiparticle density as $U/t$ grows ($N_g = 0$ and 2; $\Delta V = 0, 2$; script `make_qp_orbital_videos.py`). They are drawn on the skeleton of the dimer: the two large circles joined left to right are the two states (the direction of $R_s$, the only qp orbital that hops to the other site), and the small outer circles are their ghosts (the on-site eigenvectors orthogonal to it). Occupation as in the code (`gga_dimer.qp_state`): the lowest $N/2$ eigenvectors returned by `eigh` are filled, without fractional occupation. In the orbital movies each panel follows one orbital as $U/t$ grows (it is matched to the previous $U/t$ by maximum overlap, so it does not jump when levels cross or split; the orbitals of degenerate levels are rotated onto the previous ones); the title gives its occupation, energy and $A(\omega)$ weight. Circle area is |amplitude|$^2$ and the color is the sign (the signs of the orbitals and of the ghost circles are aligned with the previous $U/t$, so a color change is a change of the orbital, not of the phase). In the density movies the colors give the qp density per orbital (the code's $\rho$), next to the gaps, bond order, $Z$ and double occupancy (or state density at $\Delta V = 2$) as functions of $U/t$; the dashed line is $U_c/t$.

- $N_g = 0$, $\Delta V = 0$: orbitals, density
- $N_g = 0$, $\Delta V = 2$: orbitals, density
- $N_g = 2$, $\Delta V = 0$: orbitals, density
- $N_g = 2$, $\Delta V = 2$: orbitals, density

What can be seen at the critical $U_c/t$ of the weight transfer (6.56 for $\Delta V = 0$, $\simeq 8.3$ for $\Delta V = 2$):

- *Spectral gap* (first panel of the density movies, lowest positive minus highest negative energy among the levels with weight $\geq 10^{-3}$): it decreases smoothly with $U/t$ (2.0 at $U/t = 0.1$, 1.14 at 4, 0.27 at 6 for $\Delta V = 0$; 4.39, 2.0 (at $U/t = 4$) and 1.19 (at 6) for $\Delta V = 2$, $N_g = 2$) and closes at $U_c$; the next grid point has a large gap ($0.03 \to 4.94$ between $U/t = 6.5$ and 6.6 at $\Delta V = 0$; $0.001 \to 4.36$ between 8.5 and 8.6 at $\Delta V = 2$, $N_g = 2$), and it grows afterwards (at $U/t = 10$: 8.59 and 5.76). The qp gap (HOMO–LUMO) is zero from $U_c$ on; before $U_c$ it follows the spectral gap, except at small $U/t$, where weightless ghost levels lie inside the gap.
- *Orbitals*: below $U_c$ the HOMO and LUMO have amplitude on the two states and carry $A(\omega)$ weight; from $U_c$ on, the orbitals at $\epsilon = 0$ have no amplitude on the states and zero weight, and sit on ghost circles only (two of them). Which of these degenerate orbitals are occupied is the choice of `eigh`, so the occupied ghost orbitals change from frame to frame above $U_c$.
- *Bond order and double occupancy* (center–center density matrix element, per spin): both change slope at $U_c$ (at $\Delta V = 0$: bond order 0.17, 0.09, 0.08 and double occupancy 0.040, 0.016, 0.013 at $U/t = 6$, 6.5, 6.6). At $\Delta V = 2$ the bond order has a maximum (0.40) at $U/t \simeq 4.3$ and falls to 0.08 at $U_c$.
- *State density at* $\Delta V = 2$: the densities of the two states are polarized at small $U/t$ (0.945 and 0.055 at $U/t = 0.1$, 0.63 and 0.37 at 6) and become equal at $U_c$ (0.50 and 0.50, $N_g = 2$), staying equal above.
- $Z$ stays $\geq 0.975$ for all $U/t$ (no collapse of the renormalization), and at $\Delta V = 0$ the state densities are 0.50 at all $U/t$; the colors of the ghost circles are not a signature of the transition.

This is a gap closing and re-opening of the gGA solution. It is not present in the exact dimer, whose spectral gap varies smoothly with $U/t$ (Sec. 3).

### 3.4 Physical orbitals: the quasiparticle orbitals transformed by $R$

With $\Phi_{qp}$ the matrix of eigenvectors of $H_{qp}$ (dimension $2(1 + N_g)$; lowest $1 + N_g$ occupied) and $R$ the $2 \times 2(1 + N_g)$ matrix with row $s$ equal to $R_s$ on the block of site $s$, the physical orbitals are $\phi^{phys} = R\,\Phi_{qp}$, i.e. two orbitals (the rows) with coefficients $c_{R_s,n} = R_s \cdot \phi^{qp}_n|_{\text{block } s}$ over the

qp eigenorbitals. Their squared norm is $\sum_n c^2_{R_s,n} = |R_s|^2 < 1$ (e.g. 0.996 at $N_g = 2$, $\Delta V = 0$, $U/t = 4.9$). Movies: left, the coefficients of each physical orbital (one circle per qp orbital, occ/virt and $\epsilon_{qp}$ labelled, area $c^2$, color = sign, phases aligned in $U/t$); right, the sum $\sum_n c_{R_s,n}\phi^{qp}_n$ on the two physical sites, with its norm. Since $\Phi_{qp}\Phi^T_{qp} = 1$ the sum is $R^T_s$, so the final orbital sits on site $s$ and the other site is empty (localized by construction).

- $N_g = 0$, $\Delta V = 0$: physical orbitals
- $N_g = 0$, $\Delta V = 2$: physical orbitals
- $N_g = 2$, $\Delta V = 0$: physical orbitals
- $N_g = 2$, $\Delta V = 2$: physical orbitals

### 3.5 $N_g = 0$ orbitals versus the $N_g = 2$ HOMO and LUMO

**Relation between the two.** At each $U/t$ the part of the $N_g = 2$ HOMO and LUMO that sits on the two states has the same shape as the $N_g = 0$ occupied and virtual orbital: the same relative amplitudes and signs (the global signs of the $N_g = 0$ orbitals are chosen to match). The rest of the $N_g = 2$ orbitals sits on the ghosts; it takes weight away from the state part, so the weight of the $N_g = 2$ HOMO and LUMO vanishes at a lower $U_c/t$ than the $N_g = 0$ one (6.56 and 8.33 against 8.00 and 9.66 at $\Delta V = 0$ and 2). The transition does *not* take place at the same $U/t$, but the orbitals are the same.

Table 2 gives, for the $N_g = 2$ HOMO, the weight $n_s$ on the two states, the overlap between its state part and the $N_g = 0$ occupied orbital, the level energies and the $A(\omega)$ weights of both (the LUMO has the same values with the opposite energy; script `ng0_vs_ng2_orbitals.py`). The overlap is 1.000 over the whole range (0.996 at $U/t = 8$, $\Delta V = 2$, where $n_s$ is already 0.09), $n_s$ falls to zero at $U_c$, and the weight is $A \simeq 2n_s$ for $N_g = 2$ against $2Z$ for $N_g = 0$. At $\Delta V = 2$ and $U/t \lesssim 1.5$ the $N_g = 2$ HOMO and LUMO are weightless ghost levels, and the orbitals that carry the state weight are the deeper ones.

| | $N_g = 2$ HOMO: state part | | $\lvert\epsilon\rvert/t$ | | $A(\omega)$ weight | |
|---|---|---|---|---|---|---|
| $U/t$ | $n_s$ | overlap with $N_g = 0$ | $N_g = 2$ | $N_g = 0$ | $N_g = 2$ | $N_g = 0$ |
| $\Delta V = 0$ | | | | | | |
| 2 | 0.86 | 1.000 | 0.88 | 0.94 | 1.72 | 1.88 |
| 4 | 0.54 | 1.000 | 0.57 | 0.75 | 1.08 | 1.50 |
| 5 | 0.35 | 1.000 | 0.36 | 0.61 | 0.69 | 1.22 |
| 6 | 0.13 | 1.000 | 0.14 | 0.44 | 0.27 | 0.88 |
| $\Delta V = 2$ | | | | | | |
| 2 | 0.98 | 1.000 | 1.52 | 1.54 | 1.96 | 1.99 |
| 4 | 0.86 | 1.000 | 1.02 | 1.08 | 1.73 | 1.85 |
| 6 | 0.55 | 1.000 | 0.59 | 0.75 | 1.09 | 1.45 |
| 8 | 0.09 | 0.996 | 0.09 | 0.38 | 0.18 | 0.76 |

Table 2: $N_g = 2$ HOMO against the $N_g = 0$ occupied orbital: weight $n_s$ on the two states, overlap of the state part with the $N_g = 0$ orbital, level energy and $A(\omega)$ weight.

### 3.6 Ionic and covalent weights

As $U/t$ grows (the analogue of stretching $H_2$) the weight of the ionic configurations (one site doubly occupied, $H^-H^+$) is expected to go to the covalent ones (one electron on each site). The

ionic weight is $P_{\rm ion} = \langle n_{0\uparrow} n_{0\downarrow} \rangle + \langle n_{1\uparrow} n_{1\downarrow} \rangle$ and the covalent weight is $1 - P_{\rm ion}$. In gGA the double occupancies and the site occupations come from the impurity ground state of each fragment (`gga_dimer.impurity_solve`) at the converged $(R, \lambda)$, as in one iteration of the loop (at $\Delta V = 0$ they reproduce the `docc` of the solver output); ED uses the exact ground state ($\Delta V_{\rm ED} = 2\,\Delta V_{\rm gGA}$). Figure 4 gives $P_{\rm ion}$ over the whole $U/t$ grid for both $\Delta V$.

| | $\Delta V = 0$: $P_{\rm ion}$ | | | $\Delta V = 2$: $P_{\rm ion}$ | | | $\Delta V = 2$: $n$ favored site | |
|---|---|---|---|---|---|---|---|---|
| $U/t$ | ED | $N_g = 0$ | $N_g = 2$ | ED | $N_g = 0$ | $N_g = 2$ | ED | $N_g = 2$ |
| 0.1 | 0.488 | 0.494 | 0.494 | 0.896 | 0.896 | 0.896 | 1.890 | 1.890 |
| 2 | 0.276 | 0.375 | 0.374 | 0.767 | 0.781 | 0.781 | 1.755 | 1.764 |
| 4 | 0.146 | 0.250 | 0.238 | 0.469 | 0.558 | 0.557 | 1.446 | 1.517 |
| 6 | 0.084 | 0.125 | 0.079 | 0.205 | 0.321 | 0.310 | 1.178 | 1.261 |
| 7 | 0.066 | 0.062 | 0.022 | 0.138 | 0.220 | 0.193 | 1.113 | 1.154 |
| 8.5 | 0.048 | 0 | 0.015 | 0.083 | 0.089 | 0.028 | 1.061 | 1.021 |
| 10 | 0.036 | 0 | 0.010 | 0.055 | 0 | 0.017 | 1.036 | 1.011 |

Table 3: Ionic weight (ED and gGA, $N_g = 0$ and 2) and, at $\Delta V = 2$, electrons on the favored site ($n = 1$ is purely covalent).

At $\Delta V = 0$ there is no charge transfer ($n = 1$ on each site) and the ionic weight falls from 0.5, i.e. the weight moves from ionic to covalent, in ED smoothly (0.036 at $U/t = 10$). $N_g = 0$ falls linearly and vanishes at $U_c/t = 8$; it agrees with $N_g = 2$ up to $U/t \simeq 2$ and is more ionic for $2 \lesssim U/t \lesssim 6$ (0.125 against 0.079 at $U/t = 6$). gGA ($N_g = 2$) does the same but is too ionic at intermediate $U/t$ and then drops across $U_c/t \simeq 6.56$ ($0.079 \to 0.022$ between $U/t = 6$ and 7) to an almost purely covalent state (0.010 at $U/t = 10$). At $\Delta V = 2$ the state is polarized ($H^-H^+$-like, 1.89 electrons on the favored site) and becomes covalent as $U/t$ grows; gGA is more ionic than ED up to $U/t \simeq 7$ and less ionic after $U_c/t \simeq 8.3$ ($n = 1.021$ against 1.061 at $U/t = 8.5$). The density of the two states in the density movies matches the physical occupation per spin at the two points checked (0.945 at $U/t = 0.1$ and 0.630 at 6 for $\Delta V = 2$).

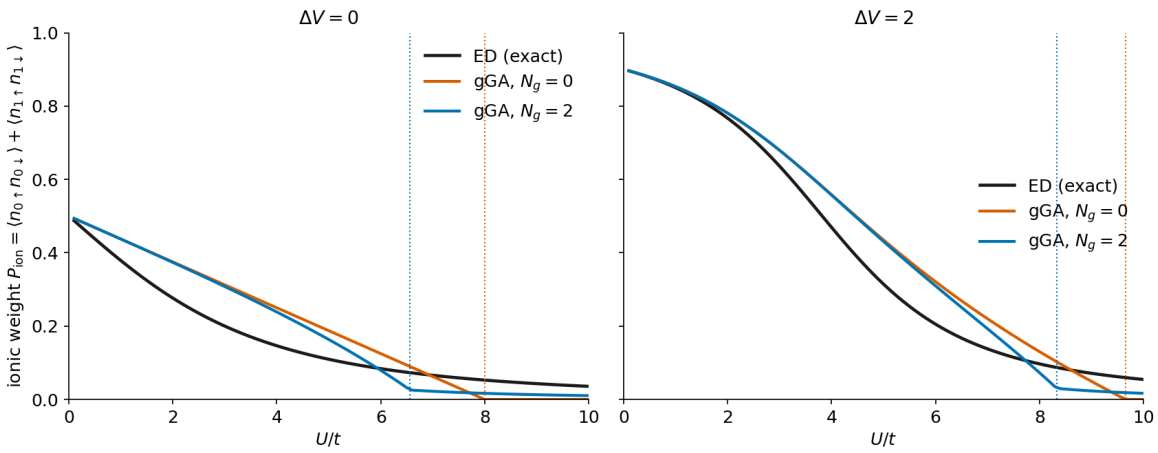



Figure 4: Ionic weight $P_{\rm ion}$ of the ground state vs. $U/t$: ED (black), gGA with $N_g = 0$ (orange) and $N_g = 2$ (blue), for $\Delta V = 0$ (left) and $\Delta V = 2$ (right); the dotted lines are $U_c/t$ of $N_g = 0$ (8.00 and 9.66) and of $N_g = 2$ (6.56 and 8.34). For $N_g = 0$ the curve is the closed form of Sec. 1 at $\Delta V = 0$ and its two-variable reduction at $\Delta V = 2$. For $N_g = 2$ and $\Delta V = 0$ the gGA values are $2\,$`docc` from the solver output; at $\Delta V = 2$ they are the impurity double occupancies $\sum_I \phi_I^T N_{dd}\, \phi_I$ of the embedding states. Script: `make_ng0_figures.py`.

**$P_{\rm ion}$ from the Gutzwiller wavefunction.** In the gGA a local expectation value in the Gutzwiller wavefunction is the one in the embedding state, so $P_{\rm ion} = \sum_I \phi_I^T N_{dd}\, \phi_I$ can be read from the $\Phi$-matrix of the direct minimization, without the impurity solve. At a solution it equals the impurity value of Fig. 4 (script `pion_check_refined.py`). The ionic weight of the explicitly built, normalized $\Psi_G$ (Sec. 6) agrees with it qualitatively.

## 4 Critical $U_c/t$ versus $\Delta V$

Figure 5 follows the transition of the gGA solution as the site splitting grows from $\Delta V = 0$ to 5 in steps of 0.2, for $N_g = 0$ and $N_g = 2$. At each $\Delta V$ the direct-minimization solution is followed in $U/t$ (steps of 0.1). For $N_g = 2$ it starts from the $\Delta V = 0$ solution and is continued in $\Delta V$, and $U_c/t$ is the value where the spectral gap (Sec. 3) goes to zero, from a linear extrapolation of the last grid points before it closes (script `Uc_from_scan.py`, data in `scan_Uc_vs_dV/`). For $N_g = 0$ the spectral function disappears when $Z \to 0$, and $U_c/t = 4 + 2\sqrt{4 + (\Delta V/t)^2}$ (Sec. 1); the extrapolation of $Z$ to zero of the numerical solutions (script `ng0_analysis.py`, data in `scan_ng0/`) agrees with it to $1.3 \times 10^{-3}$ at all $\Delta V$. The values at $\Delta V = 0$ and 2 reproduce those of Secs. 1 and 3 (8.00 and 9.66 for $N_g = 0$; 6.56 and 8.33 for $N_g = 2$).

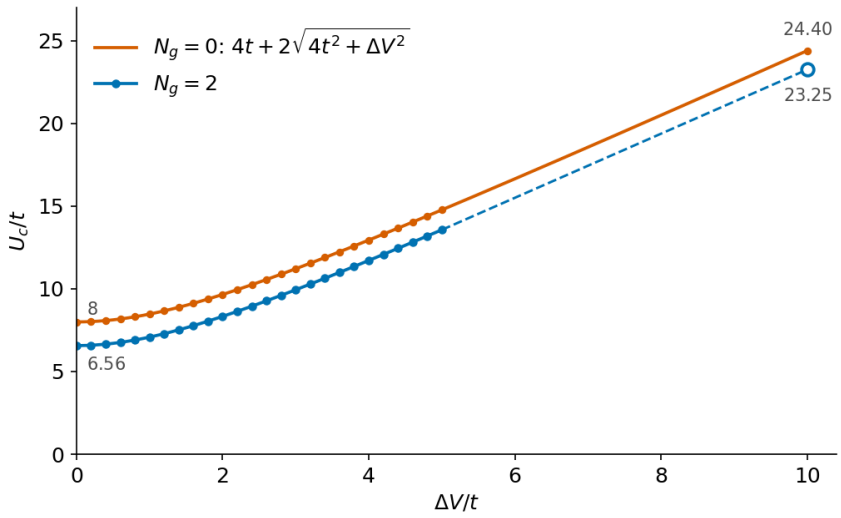



Figure 5: Critical $U_c/t$ versus $\Delta V$ for $N_g = 0$ (orange; curve: closed form, dots: numerical solutions) and $N_g = 2$ (blue); the open circle is the single point at $\Delta V = 10$ of $N_g = 2$, computed separately, and the dashed line is a straight segment joining it to $\Delta V = 5$ (no points in between). Script: `make_ng0_figures.py`.

| $\Delta V/t$ | 0 | 1 | 2 | 3 | 5 | 10 |
|---|---|---|---|---|---|---|
| $U_c/t$, $N_g = 0$ | 8.00 | 8.47 | 9.66 | 11.21 | 14.77 | 24.40 |
| $U_c/t$, $N_g = 2$ | 6.56 | 7.08 | 8.33 | 9.94 | 13.57 | 23.25 |

Table 4: Critical $U_c/t$ for $N_g = 0$ (closed form) and $N_g = 2$.

For both $N_g$, $U_c/t$ increases monotonically with $\Delta V$: it is flat at small $\Delta V$ ($N_g = 2$: 6.56, 6.59, 6.65 at $\Delta V = 0$, 0.2, 0.4) and grows almost linearly for $\Delta V \gtrsim 3$ (slope $\simeq 1.8$ for both; mean slopes 1.93 for $N_g = 0$ and 1.94 for $N_g = 2$ between $\Delta V = 5$ and 10; the $N_g = 0$ closed form tends to slope 2). $N_g = 0$ lies above $N_g = 2$ at all $\Delta V$, by 1.44 at $\Delta V = 0$ and 1.14 at $\Delta V = 10$: the ghost orbitals lower $U_c$.

## 5 Discussion: $\Delta V = 0$ vs. $\Delta V = 2$

**Weak coupling: $\Delta V$ dominates.** At small $U/t$ the two scans differ substantially. At $U/t = 0.1$, $E_{\text{var}}/t = -2.0503$ for $\Delta V = 0$ but $-4.4823$ for $\Delta V = 2$. This is the expected $U \to 0$ limit: with $\Delta V = 0$ the dimer's noninteracting energy is $-2t = -2$ plus a small $O(U)$ correction, while at $\Delta V = 2$ the site-splitting opens a one-body level gap and the $U \to 0$ energy approaches the closed form $E = -2\sqrt{t^2 + \Delta V^2} = -2\sqrt{5} \approx -4.472$. The gap between the two curves is largest here, $\Delta E \approx 2.43$, because $\Delta V$ acts as an unscreened one-body potential that the (still weak) correlations barely renormalize.

**Strong coupling: the curves converge.** As $U/t$ grows the two energies draw close together: at $U/t = 10$, $E_{\text{var}}/t = -10.101$ ($\Delta V = 0$) vs. $-10.120$ ($\Delta V = 2$), a difference of only $\approx 0.02$ – two orders of magnitude smaller than at weak coupling. In the Mott-like regime the total energy is

governed almost entirely by $U$ (both sites approach single occupancy), and the fixed $\Delta V = 2$ site-splitting becomes a small perturbation on top of the dominant Hubbard physics. The quasiparticle spectra (3D plots, Fig. 3) show the same trend: the $\Delta V = 2$ branches, well separated at small $U$, bunch together with the $\Delta V = 0$-like structure as $U/t$ increases.

**Level crossing at $\Delta V = 2$.** The $\Delta V = 2$ quasiparticle energies show a sharp rearrangement between $U/t = 0.6$ and $U/t = 0.7$ (e.g. $N_g = 2$: $\epsilon^{qp} \approx -2.00 \to -3.12$), visible as a kink in the 3D plots; this is the level crossing where the $\Delta V$-induced one-body gap stops being the dominant scale and the Hubbard repulsion takes over.

**Ghost multiplicity ($N_g = 0$ vs. $N_g = 2$).** The ghost orbitals lower the energy only at larger $U/t$: $E(N_g{=}0) - E(N_g{=}2) < 10^{-3}$ up to $U/t = 2.3$ at $\Delta V = 0$ and 4.3 at $\Delta V = 2$, and at most 0.13 in the scanned range (Sec. 2), still well below the error to the exact energy at large $U/t$ ($\approx 0.3$–$0.5$). The ghosts change the transition more than the energy: $U_c/t$ goes from 8 ($N_g = 0$) to 6.56 ($N_g = 2$) at $\Delta V = 0$ and the transition is no longer a continuous vanishing of $Z$ (the $N_g = 0$ spectral function disappears with $Z$, the $N_g = 2$ one keeps $Z \geq 0.975$ and loses its inner peak, Sec. 3).

## 6 The Gutzwiller wave function itself

gGA never forms the state it describes. Here it is built explicitly from the converged solutions (script `gutzwiller_psi.py`; scans `psi_scan.py`, data in `psi_scan/`), to read off the ionic and covalent content, the energy and the overlap with the exact ground state (FCI, from `exact_dimer_deltaV.py`).

**Construction.** For each fragment, $|\Psi\rangle = P_0 P_1 |\Psi_{\rm qp}\rangle$, with $|\Psi_{\rm qp}\rangle$ the Slater determinant of the occupied levels of $H_{qp}$ and $P_i = \sum_{\Gamma n} \lambda_{\Gamma n} |\Gamma\rangle\langle n|$ mapping the qp Fock space of the fragment ($4^{N_{\rm eff}}$ states, $N_{\rm eff} = 1 + N_g$) onto its four physical configurations $\Gamma$. The matrix is $\lambda = \tilde{\Phi}\, P^{0\,-1/2}$, with $P^0$ the local density matrix of $\Psi_{\rm qp}$ and $\tilde{\Phi}$ the embedding state $\Phi$ with the bath particle-hole transformed back, $\tilde{\Phi} = \Phi\, U_{\rm PH}$, $U_{\rm PH}|n\rangle = (-1)^{N(N-1)/2} f_{j_1} \cdots f_{j_N} |{\rm full}\rangle$. The phase fixes the relative sign of the covalent and ionic amplitudes (the other choice gives $E_{\rm kin} > 0$). Checks: $R = S^{-1}{\rm Tr}[\tilde{\Phi}^T c^\dagger \tilde{\Phi} f]$ reproduces the $R$ of the solution to $10^{-7}$ (fully for the non-impurity points), and the weight of $\tilde{\Phi}$ outside the support of $P^0$ is $\leq 7 \times 10^{-8}$. $\Phi$ is the saved $\phi_i$ for non-impurity solutions (`z.npy`) and the ground state of $H_{\rm imp}$ for impurity solutions. $N_g = 2$, $\Delta V = 2$ uses the non-impurity scan (`scan_ng2_not_imp_dV2`): the impurity-solver points (`scan_ng2_dV2_confirmed`) have the same energies, but at $U/t = 0.6, 0.7, 9.9, 10$ the Fermi level is degenerate or the ghosts are inert, $\Psi_{\rm qp}$ does not match $\Phi$, and $\Psi$ fails there.

**Normalization and energy.** The gGA constraints give $\langle\Psi|\Psi\rangle = 1$ only in infinite dimensions; here it grows from 1 to $\simeq 2.8$ with $U/t$ (Tables 5, 6; 0.29 for $N_g = 2$, $\Delta V = 2$ at small $U/t$, where the ghosts are inert and the value depends on the gauge). The origin of this norm is worked out in the note `psi_norm.tex` (`psi_norm.pdf`): $\langle\Psi|\Psi\rangle = \langle\Psi_0|G_0 G_1|\Psi_0\rangle$ with $G_i = P_i^\dagger P_i$, each factor has expectation value 1 by the gGA constraints, and the excess is the correlation of $G_0$ and $G_1$ in $\Psi_0$; for $N_g = 0$, $\Delta V = 0$ it is $1 + (U/8t)^2$ in closed form, which reproduces the $N_g = 0$ column of the tables. All quantities below use the normalized $\Psi$. The energy $E_\Psi = \langle\Psi|H|\Psi\rangle$ is variational and far better than the gGA one: for $N_g = 2$ the error is $\leq 8.5 \times 10^{-4}$ over the whole scan (against up to 0.4 for gGA) and the overlap with FCI is $\geq 0.9999$. For $N_g = 0$ it is 0.03 at $U/t = 4$ and 0.3 at 7 (overlap 0.98 at $U_c$); $N_g = 0$ at $\Delta V = 0$ is the ordinary Gutzwiller wave function, and gGA's $E$ is its Gutzwiller-approximation value. Above $U_c$ of $N_g = 0$ ($Z = 0$, $R = 0$) the solution does not

determine $\Psi_{\rm qp}$: the values marked $^\dagger$ (dashed in Fig. 6) depend on the arbitrary $\Psi_{\rm qp}$ stored by the solver (e.g. $E_\Psi = -U$ and $P_{\rm ion} = 0$ exactly) and are not a result.

| | $N_g = 0$ | | | | $N_g = 2$ | | | |
|---|---|---|---|---|---|---|---|---|
| $U/t$ | $\langle\Psi\|\Psi\rangle$ | $E_{\rm gGA} - E$ | $E_\Psi - E$ | $1 - \|\langle\Psi\|{\rm FCI}\rangle\|$ | $\langle\Psi\|\Psi\rangle$ | $E_{\rm gGA} - E$ | $E_\Psi - E$ | $1 - \|\langle\Psi\|{\rm FCI}\rangle\|$ |
| 0.1 | 1.000 | 0.000 | 0 | 0 | 1.000 | 0.000 | 0 | 0 |
| 2 | 1.062 | 0.111 | $7.7\times10^{-4}$ | $8.7\times10^{-5}$ | 1.153 | 0.110 | $5.4\times10^{-5}$ | $6.0\times10^{-6}$ |
| 4 | 1.250 | 0.328 | $2.8\times10^{-2}$ | $2.5\times10^{-3}$ | 1.758 | 0.317 | $1.5\times10^{-5}$ | $1.3\times10^{-6}$ |
| 6 | 1.562 | 0.481 | $1.7\times10^{-1}$ | $1.2\times10^{-2}$ | 2.647 | 0.418 | $2.3\times10^{-4}$ | $1.6\times10^{-5}$ |
| 7 | 1.766 | 0.500 | $3.0\times10^{-1}$ | $1.9\times10^{-2}$ | 2.731 | 0.384 | $2.4\times10^{-4}$ | $1.5\times10^{-5}$ |
| 8.5 | $2.000^\dagger$ | $0.447^\dagger$ | $4.5\times10^{-1\dagger}$ | $2.4\times10^{-2\dagger}$ | 2.466 | 0.327 | $1.2\times10^{-4}$ | $6.3\times10^{-6}$ |
| 10 | $2.000^\dagger$ | $0.385^\dagger$ | $3.9\times10^{-1\dagger}$ | $1.8\times10^{-2\dagger}$ | 2.325 | 0.284 | $6.1\times10^{-5}$ | $2.8\times10^{-6}$ |

Table 5: $\Delta V = 0$: norm, error of the gGA energy $E_{\rm gGA} - E_{\rm FCI}$ and of $E_\Psi - E_{\rm FCI}$, and overlap deficit of $\Psi$ with the FCI ground state. $^\dagger$: above $U_c$ of $N_g = 0$, $\Psi_{\rm qp}$ is not determined.

| | $N_g = 0$ | | | | $N_g = 2$ | | | |
|---|---|---|---|---|---|---|---|---|
| $U/t$ | $\langle\Psi\|\Psi\rangle$ | $E_{\rm gGA} - E$ | $E_\Psi - E$ | $1 - \|\langle\Psi\|{\rm FCI}\rangle\|$ | $\langle\Psi\|\Psi\rangle$ | $E_{\rm gGA} - E$ | $E_\Psi - E$ | $1 - \|\langle\Psi\|{\rm FCI}\rangle\|$ |
| 0.1 | 1.000 | 0.000 | 0 | 0 | 0.294 | 0.000 | 0 | 0 |
| 2 | 1.003 | 0.008 | $2.1\times10^{-4}$ | $2.0\times10^{-5}$ | 0.294 | 0.008 | $2.1\times10^{-4}$ | $2.0\times10^{-5}$ |
| 4 | 1.043 | 0.101 | $6.8\times10^{-3}$ | $5.9\times10^{-4}$ | 1.106 | 0.100 | $3.1\times10^{-4}$ | $5.0\times10^{-5}$ |
| 6 | 1.216 | 0.329 | $3.4\times10^{-2}$ | $1.6\times10^{-3}$ | 1.651 | 0.319 | $1.2\times10^{-4}$ | $1.6\times10^{-5}$ |
| 7 | 1.372 | 0.430 | $7.9\times10^{-2}$ | $4.8\times10^{-3}$ | 2.099 | 0.402 | $1.7\times10^{-4}$ | $2.0\times10^{-5}$ |
| 8.5 | 1.689 | 0.499 | $2.4\times10^{-1}$ | $1.6\times10^{-2}$ | 2.700 | 0.397 | $6.3\times10^{-4}$ | $5.7\times10^{-5}$ |
| 10 | $2.060^\dagger$ | $0.449^\dagger$ | $4.5\times10^{-1\dagger}$ | $5.0\times10^{-1\dagger}$ | 2.426 | 0.328 | $2.2\times10^{-4}$ | $1.6\times10^{-5}$ |

Table 6: As Table 5 for $\Delta V = 2$ ($N_g = 2$ from the non-impurity scan).

**Ionic → covalent.** Table 7 compares $P_{\rm ion}$ of ED, of $\Psi$ ($\Psi_{N_g=0}$, $\Psi_{N_g=2}$) and of the fragment ground states used so far (frag, Sec. 3). The fragments are too ionic up to $U/t \simeq 5$ (${\rm frag}_{N_g=2}$: 0.238 against 0.146 at $U/t = 4$) and too covalent from 6 on (0.079 and 0.022 against 0.084 and 0.066 at 6 and 7), while $\Psi_{N_g=2}$ follows ED within 0.006 at $\Delta V = 0$ and 0.011 at $\Delta V = 2$ (0.1453 against 0.1464 at $U/t = 4$) and keeps the slow decrease above $U_c$ (0.037 against 0.036 at 10). $\Psi_{N_g=0}$ is too covalent: 0.100 at $U/t = 4$ and 0.020 at 6, going to 0 at $U_c$.

| | $\Delta V = 0$: $P_{\rm ion}$ | | | | | $\Delta V = 2$: $P_{\rm ion}$ | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $U/t$ | ED | $\Psi_{N_g=0}$ | $\Psi_{N_g=2}$ | ${\rm frag}_{N_g=0}$ | ${\rm frag}_{N_g=2}$ | ED | $\Psi_{N_g=0}$ | $\Psi_{N_g=2}$ | ${\rm frag}_{N_g=0}$ | ${\rm frag}_{N_g=2}$ |
| 0.1 | 0.4875 | 0.4875 | 0.4875 | 0.4938 | 0.4937 | 0.8961 | 0.8961 | 0.8961 | 0.8962 | 0.8962 |
| 2 | 0.2764 | 0.2647 | 0.2733 | 0.3750 | 0.3737 | 0.7669 | 0.7702 | 0.7702 | 0.7808 | 0.7808 |
| 4 | 0.1464 | 0.1000 | 0.1453 | 0.2500 | 0.2385 | 0.4693 | 0.4888 | 0.4783 | 0.5583 | 0.5572 |
| 6 | 0.0840 | 0.0200 | 0.0871 | 0.1250 | 0.0794 | 0.2052 | 0.1822 | 0.2093 | 0.3211 | 0.3096 |
| 7 | 0.0659 | 0.0044 | 0.0686 | 0.0625 | 0.0223 | 0.1379 | 0.0847 | 0.1419 | 0.2201 | 0.1928 |
| 8.5 | 0.0476 | $0.0000^\dagger$ | 0.0491 | $0.0000^\dagger$ | 0.0147 | 0.0827 | 0.0128 | 0.0884 | 0.0889 | 0.0282 |
| 10 | 0.0358 | $0.0000^\dagger$ | 0.0367 | $0.0000^\dagger$ | 0.0104 | 0.0545 | $0.0000^\dagger$ | 0.0570 | $0.0000^\dagger$ | 0.0167 |

Table 7: Ionic weight $P_{\rm ion}$: ED, $\Psi_{N_g=0}$, $\Psi_{N_g=2}$ and fragment ground states (frag). $^\dagger$: $\Psi_{\rm qp}$ not determined (above $U_c$ of $N_g = 0$).

At $\Delta V = 0$, $|\Psi\rangle = \cos\theta|\mathrm{cov}\rangle + \sin\theta|\mathrm{ion}\rangle$ with the amplitudes of Table 8. $\Psi_{N_g=2}$ has the exact amplitudes to 0.01 over the scan and the same relative sign. The last columns give $(\cos\theta + \sin\theta)^2$, the weight of the bonding (inner) cation state in the exact interference relation, evaluated with each amplitude pair (the final states are the exact cation eigenstates, so this is a property of the ground state, not a gGA spectrum): 1.7048 ($\Psi_{N_g=2}$), 1.7071 (ED) and 1.6000 ($\Psi_{N_g=0}$) at $U/t = 4$ and 1.5640 ($\Psi_{N_g=2}$), 1.5547 (ED), 1.2800 ($\Psi_{N_g=0}$) at 6; the lowest peak of the gGA spectrum has 1.08 at $U/t = 4$ and 0.27 at 6. So the ionic→covalent transfer is reproduced by $\Psi_{N_g=2}$ but not by the gGA spectrum: the weight carried by $R$ and $H_{qp}$ does not follow the composition of the state it comes from.

| | ED | | $\Psi_{N_g=2}$ | | $\Psi_{N_g=0}$ | | $(\cos\theta + \sin\theta)^2$ | | |
|---|---|---|---|---|---|---|---|---|---|
| $U/t$ | $\cos\theta$ | $\sin\theta$ | $\cos\theta$ | $\sin\theta$ | $\cos\theta$ | $\sin\theta$ | ED | $\Psi_{N_g=2}$ | $\Psi_{N_g=0}$ |
| 0.1 | 0.7159 | 0.6982 | 0.7159 | 0.6982 | 0.7159 | 0.6982 | 1.9997 | 1.9997 | 1.9997 |
| 2 | 0.8507 | 0.5257 | 0.8525 | 0.5228 | 0.8575 | 0.5145 | 1.8944 | 1.8913 | 1.8824 |
| 4 | 0.9239 | 0.3827 | 0.9245 | 0.3812 | 0.9487 | 0.3162 | 1.7071 | 1.7048 | 1.6000 |
| 6 | 0.9571 | 0.2898 | 0.9554 | 0.2952 | 0.9899 | 0.1414 | 1.5547 | 1.5640 | 1.2800 |
| 7 | 0.9665 | 0.2567 | 0.9651 | 0.2619 | 0.9978 | 0.0665 | 1.4961 | 1.5056 | 1.1327 |
| 8.5 | 0.9759 | 0.2182 | 0.9751 | 0.2216 | 1.0000$^\dagger$ | 0.0000 | 1.4258 | 1.4322 | 1.0000$^\dagger$ |
| 10 | 0.9820 | 0.1891 | 0.9815 | 0.1914 | 1.0000$^\dagger$ | 0.0000 | 1.3714 | 1.3758 | 1.0000$^\dagger$ |

Table 8: $\Delta V = 0$: covalent and ionic amplitudes of the exact ground state and of $\Psi_{N_g=2}$, $\Psi_{N_g=0}$ (normalized, global sign with $\cos\theta > 0$), and $(\cos\theta + \sin\theta)^2$. $^\dagger$: $\Psi_{\mathrm{qp}}$ not determined (above $U_c$).

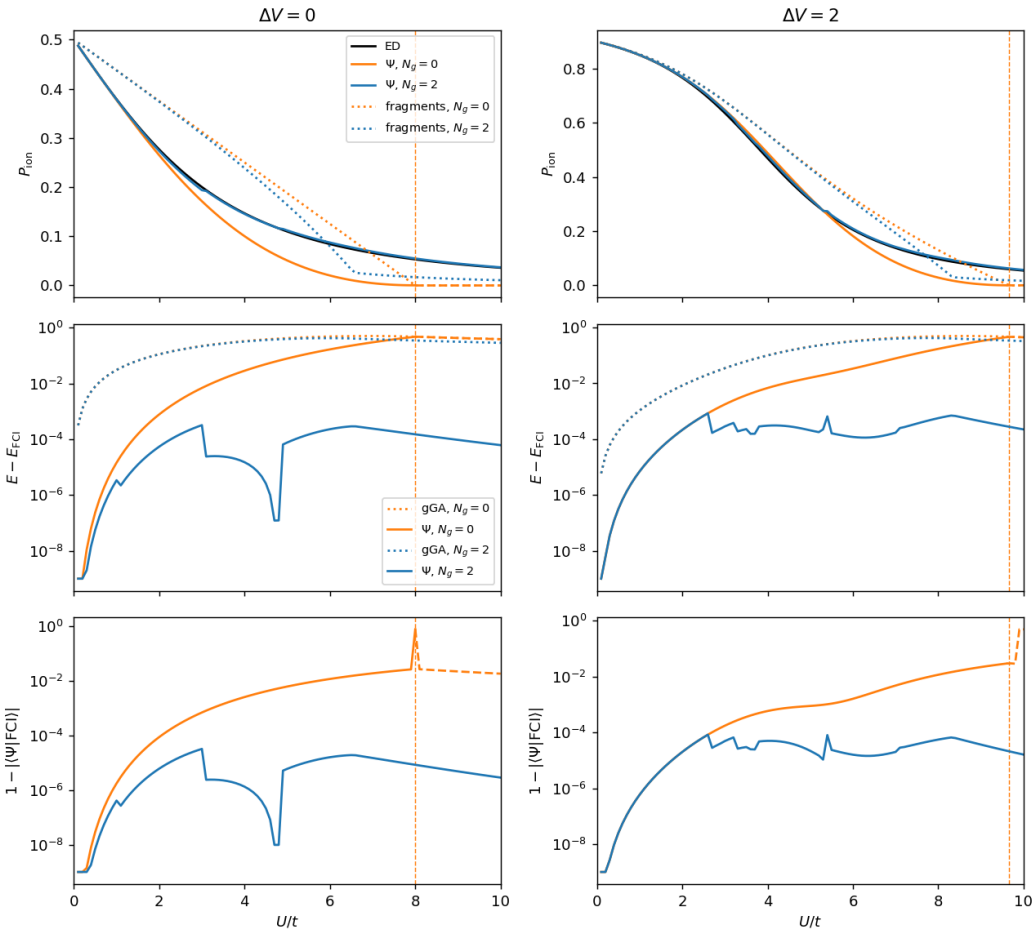



Figure 6: Whole scans. Top: $P_{\rm ion}$ (ED, Ψ, fragments). Middle: $E - E_{\rm FCI}$ for gGA (dotted) and Ψ. Bottom: $1 - |\langle\Psi|{\rm FCI}\rangle|$. $N_g = 0$ is dashed above its $U_c/t$ (8.00, 9.66; vertical lines), where $\Psi_{\rm qp}$ is not determined.

### 6.1 Ionic weight of Ψ at fixed $|R|$

To see how the ionic weight of Ψ depends on $R$, the gGA solution is recomputed with $R$ constrained, so that the state stays a solution of the gGA equations for that value of $R$. The ghost-GA functional of the direct minimization (Sec. 6; $\Delta V = 0$) is minimized directly with its constraints (class `NotImpDimer` of `not_imp_gga_dimer.py`, SLSQP), with no impurity problem solved and no self-consistency loop, and with the additional equality constraint $R_i^T R_i = s^2 |R_i^{\rm sol}|^2$ on both fragments, where $R^{\rm sol}$ is the unmodified solution ($s = 1$). Starting from it, $s$ is changed step by step, each point starting from the previous one (largest step 0.05 for $N_g = 0$ and 0.02 for $N_g = 2$, halved when a step does not converge), and Ψ is built from every converged point as in Sec. 6 (script `psi_fixedR.py`; data in `psi_scan/Rfix_*`, solutions in `Rfixed/`). The constrained minimization (functional, constraint on $|R_i|^2$, SLSQP, continuation in $s$) is described step by step in the separate note `fixedR_constraint.tex` (`fixedR_constraint.pdf`).

$N_g = 0$. $R_i$ is one number, $R_i = \sqrt{Z_i}$, so fixing $R_i^2$ fixes $R_i$ itself and there is no norm to keep: $s$ changes $R$, and the reference value $|R^{\rm sol}| = 0.968, 0.866, 0.781, 0.661, 0.484$ at $U/t = 2, 4, 5, 6, 7$ (it

vanishes at $U_c/t = 8$) is smaller than the 0.99–1.00 of $N_g = 2$ and decreases with $U/t$. $P_{\rm ion}$ rises with $R$ (Fig. 7, left; Table 9): at $U/t = 4$ it is 0.070, 0.100, 0.145 at $s = 0.95$, 1, 1.05 and 0.026 at $s = 0.80$, and it goes to 0.003 at $s = 0.5$. The curves are close to straight lines on the logarithmic scale of the figure, for all $U/t$ computed $(2, 4, 5, 6, 7)$. The values reached with $s > 1$ are limited: $s = 1.1$ for $U/t \geq 4$ (the largest value tried) and only $s = 1.033$ at $U/t = 2$, where $P_{\rm ion}$ is already 0.49. The gGA energy has its minimum at $s = 1$, as it must, and rises on both sides: at $U/t = 4$ by 0.009 and 0.014 for $s = 0.95$ and 1.05 and by 0.098 at $s = 0.80$.

| $U/t$ | $P_{\rm ion}(1.05)$ | $P_{\rm ion}(1)$ | $P_{\rm ion}(0.95)$ | $P_{\rm ion}(0.90)$ | $E_{\rm var}(0.90) - E_{\rm var}(1)$ |
|---|---|---|---|---|---|
| 2 | – | 0.265 | 0.160 | 0.105 | 0.116 |
| 4 | 0.145 | 0.100 | 0.070 | 0.050 | 0.032 |
| 5 | 0.069 | 0.051 | 0.037 | 0.028 | 0.015 |
| 6 | 0.026 | 0.020 | 0.015 | 0.012 | 0.006 |
| 7 | 0.006 | 0.004 | 0.004 | 0.003 | 0.001 |

Table 9: $N_g = 0$, $\Delta V = 0$: ionic weight of $\Psi$ and gGA energy change for $|R_i| = s\,|R_i^{\rm sol}|$ (self-consistent). At $U/t = 2$, $s = 1.05$ is not reached.

**$N_g = 2$.** $R_i$ has three components. Only $|R_i|$ can be fixed: the functional is invariant under the orthogonal rotations of the qp basis of each fragment, which turn $R_i$ into another vector of the same length, so fixing $R_i$ with $|R_i|$ unchanged returns the same energy and the same $\Psi$ (checked at $U/t = 4$: $E_{\rm var} = -4.511136$ and $P_{\rm ion} = 0.1453$ for all $s$). The values $s > 1$ could not be reached ($U/t = 4$, $s = 1.1$: the minimization does not converge), so $s \leq 1$ is studied. Lowering $|R|$ makes $\Psi$ more covalent, as for $N_g = 0$ (Fig. 7, right; Table 10): $P_{\rm ion}$ decreases with $s$ for $U/t = 2, 4, 5, 6, 7, 8$, from 0.273 to 0.059 at $U/t = 2$, from 0.087 to 0.036 at $U/t = 6$ and from 0.055 to 0.024 at $U/t = 8$, between $s = 1$ and $s = 0.80$. The decrease is steady for $U/t = 2, 6, 7, 8$. At $U/t = 4$ it is $0.145 \to 0.063$ between $s = 1$ and 0.90, then there is a small increase (0.065 at $s = 0.88$, 0.095 at 0.86) and the decrease resumes (0.059 at $s = 0.80$); the overlap of $\Psi$ with the FCI ground state is 0.997 at $s = 0.86$ and $E_{\rm var}$ has no feature there. At $U/t = 5$ the last point ($s = 0.80$, 0.027) lies below the trend of the others. The change of $P_{\rm ion}$ is large for a small change of the gGA energy: at $U/t = 4$ a change of $|R|$ by 2% ($s = 0.98$) takes $P_{\rm ion}$ from 0.145 to 0.100 and changes $E_{\rm var}$ by 0.0015 (bottom right panel of Fig. 7; $E_{\rm var}(0.80) - E_{\rm var}(1) = 0.03$–$0.04$ for $U/t \geq 4$ and 0.25 at $U/t = 2$).

| $U/t$ | $P_{\rm ion}(s = 1)$ | $P_{\rm ion}(s = 0.96)$ | $P_{\rm ion}(s = 0.80)$ | $E_{\rm var}(0.80) - E_{\rm var}(1)$ |
|---|---|---|---|---|
| 2 | 0.273 | 0.240 | 0.059 | 0.246 |
| 4 | 0.145 | 0.078 | 0.059 | 0.032 |
| 5 | 0.112 | 0.104 | 0.027 | 0.026 |
| 6 | 0.087 | 0.075 | 0.036 | 0.036 |
| 7 | 0.069 | 0.056 | 0.030 | 0.038 |
| 8 | 0.055 | 0.045 | 0.024 | 0.035 |

Table 10: $N_g = 2$, $\Delta V = 0$: ionic weight of $\Psi$ and gGA energy change when $|R_i|$ is reduced to $s\,|R_i^{\rm sol}|$ at a self-consistent point.

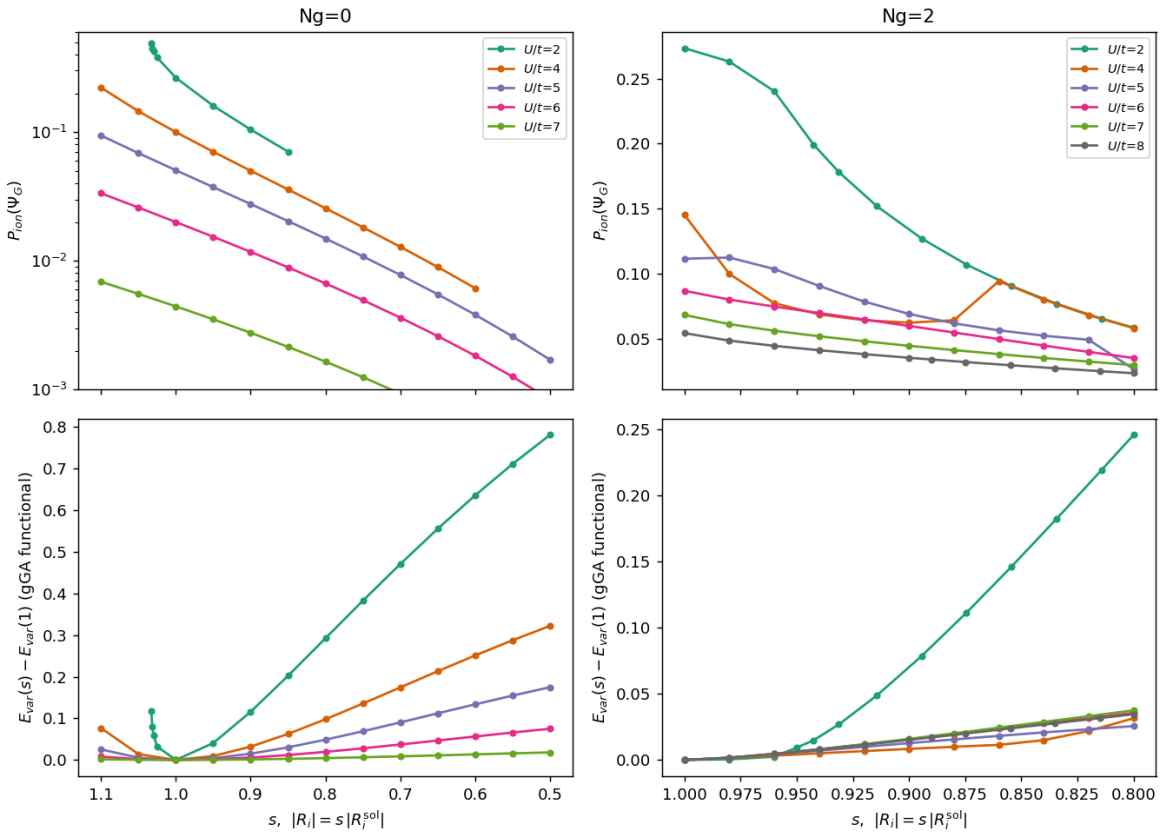



Figure 7: $\Delta V = 0$, $|R_i| = s\,|R_i^{\rm sol}|$ fixed self-consistently ($s = 1$: the solution). Left: $N_g = 0$ ($P_{\rm ion}$ on a logarithmic scale). Right: $N_g = 2$. Top: $P_{\rm ion}$ of $\Psi$. Bottom: change of the gGA energy functional. Points whose overlap with the FCI ground state is below 0.95 are left out of the top panels ($N_g = 0$: $U/t = 2$, $s \le 0.80$; $U/t = 4$, $s \le 0.55$; $N_g = 2$: $U/t = 2$, $s = 0.95$). Script `plot_fixedR.py`.

**Main conclusions of the fixed-$|R|$ test** ($\Delta V = 0$, self-consistent, $|R_i| = s\,|R_i^{\rm sol}|$).

- Lowering $|R|$ makes $\Psi$ more covalent and raising it more ionic, for $N_g = 0$ ($U/t = 2, 4, 5, 6, 7$) and, below the solution, for $N_g = 2$ ($U/t = 2, 4, 5, 6, 7, 8$).
- $N_g = 0$: $P_{\rm ion}$ is a smooth, nearly exponential function of $s$; at $U/t = 4$ it is 0.070, 0.100, 0.145 at $s = 0.95$, 1, 1.05. $R$ is the only quantity that changes ($|R^{\rm sol}| = 0.97 \to 0.48$ for $U/t = 2 \to 7$).
- $N_g = 2$: only $|R_i|$ is a meaningful variable (the direction of $R_i$ is a symmetry of the functional); $|R^{\rm sol}| \simeq 1$ at all $U/t$. A change of $|R|$ by 2% changes $P_{\rm ion}$ at $U/t = 4$ from 0.145 to 0.100, while the gGA energy changes by 0.0015.
- $N_g = 2$, $U/t = 4$: the decrease has a small bump near $s = 0.86$–$0.88$ (0.063 $\to$ 0.095), cause not established; at $U/t = 5$ the last point ($s = 0.80$) lies below the trend.
- Not covered: $s > 1$ for $N_g = 2$ (not reachable), $\Delta V \neq 0$, and $U/t$ outside the values listed.